

altairsim — CP/M 2.2 on an Altair 88-HDSK hard disk

2026-07-25 · 849e344

```
altairsim hdsk.toml

HDBL 2.00
LOADING FROM 0

48K CP/M 2.2b v1.6
For MITS 88-HDSK

A0>DIR
```

Mike Douglas's **CP/M 2.2 for the 88-HDSK** (the `HDCPM22v16` build for a 48K machine), booted by the **HDBL** PROM at `FC00` — `RUN FC00` is the machine file's whole startup, because on a real machine that was `EXAMINE FC00` and `RUN`. HDBL reads the disk's Pack Descriptor Page, loads the boot pages, and jumps into CP/M.

The **88-HDSK "Datakeeper"** is not a floppy card. It is an outboard controller (its own 8X300 processor and Pertec drive) that plugs into the Altair through an 88-4PIO at ports `A0h – A7h` and moves whole 256-byte sectors for you over a command/handshake protocol. The image is one platter: 406 cylinders × 2 sides × 24 sectors × 256 bytes = 4,988,928 bytes. See `docs/boards/mits-88hdsk.md` and `reference/88-HDSK.md`.

`^E` (ATTN) takes the keyboard back to the monitor at any point; `RUN` resumes. `^C` belongs to CP/M (it is warm boot) and CP/M gets it.

The files

File	What it is
<code>hdsk.toml</code>	The machine: <code>base = "default"</code> plus the 88-HDSK controller, the HDBL PROM at <code>FC00</code> , and the platter in drive 0.
<code>HDCPM22v16-48K.DSK</code>	The bootable system platter, built for a 48K machine.

There is no undo. Drive 0 is mounted read/write because that is what a real machine is, and CP/M writes to `A:` for anything you create. In a clone `git checkout` puts the image back; in

the package you were handed, nothing does. Copy it first if you are about to test writes in anger, or add `readonly = true` to the drive in `hdsk.toml`.

`BOOT.ASM`, `BIOS.ASM` and the original `HDBL.ASM` — the authoritative sources this was built and verified against — live at deramp.com (Altair hard-disk software and ROMs). They are source rather than product, so they are not shipped here.